

Design Patterns

Model View Controller (MVC)

The application follows an MVC-inspired architecture to separate concerns between user interaction, business logic, and data representation, improving maintainability and enabling future UI expansion.

Façade Pattern — Controller

The Controller class implements the Facade pattern by providing a simplified interface to complex subsystems such as persistence, service tracking, and vehicle state management.

Repository Pattern — DataHandler

The DataHandler class follows the Repository pattern by abstracting data storage and retrieval logic, allowing the rest of the application to remain independent of the underlying persistence mechanism.

Data Transfer Object (DTO) — ServiceStatus

ServiceStatus is implemented as a Data Transfer Object to safely communicate service evaluation results from the controller to the view layer.